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Assignment 01 – Linux Basic Command

Section A: Navigation

1 .Create a directory named linux_practice.
Command: mkdir linux_practice
I used the mkdir command to create a new directory named linux_practice in the Linux file system.

2. Go inside that directory.
Command : - cd linux_practice
I used the cd command to move inside the linux_practice directory.

3. Create another directory called test_folder.
Command : - mkdir test_folder
I used the mkdir command again to create another directory named test_folder inside the linux_practice directory.

4. Move back to the parent directory.
Command : - cd ..
I used the cd .. command to move back to the parent directory.

5. Display your current working directory
Command: - pwd
I used the pwd command to display the current working directory path.

6. List all files and directories.
Command : - ls
I used the ls command to list all files and directories in the current directory.

7. Find help for the ls command.
Command : - ls –help
I used the ls --help command to view the help information and available options for the `ls` command

Section B: File & Directory Operations

1. Create a file named student.txt
Command: - touch student.txt
I used the touch command to create a new empty file named student.txt in the current directory.

2. Create a directory called data
Command: - mkdir data
I used the mkdir command to create a new directory named data.

3. Copy student.txt inside data
Command: - cp student.txt data/
I used the cp command to copy the file **student.txt** into the **data** directory.

4. Rename student.txt to info.txt
Command: -mv student.txt info.txt
I used the mv command to rename the file **student.txt** to **info.txt**.

5. Delete the file info.txt
Command: - rm info.txt
I used the rm command to delete the file **info.txt** from the directory.

6. Remove the data directory using rmdir
Command:- rmdir data
I used the `rmdir` command to remove the **data** directory. This command works only if the directory is empty.

7. Remove the data directory using rm –rf
Command:- rm -rf data
I used the `rm -rf` command to remove the **data** directory forcefully along with all its contents.

```
2026-03-14  15:49.22  /home/mobaxterm  mkdir linux_practice
                                                                     ✔
2026-03-14  16:07.12  /home/mobaxterm  ls
Desktop  LauncherFolder  MyDocuments  README.txt  linux_practice
                                                                     ✔
2026-03-14  16:07.14  /home/mobaxterm  cd linux_practice/
                                                                     ✔
2026-03-14  16:07.40  /home/mobaxterm/linux_practice  mkdir test_folder
                                                                     ✔
2026-03-14  16:08.02  /home/mobaxterm/linux_practice  cd ..
                                                                     ✔
2026-03-14  16:08.45  /home/mobaxterm  pwd
/home/mobaxterm
                                                                     ✔
2026-03-14  16:09.02  /home/mobaxterm  ls --help
Toybox 0.8.10 multicall binary (see toybox --help)

usage: ls [-1ACFHLNRSUXZabcdfghilmnopqrstuwx] [--color[=auto]] [FILE...]

List files

what to show:
-A  all files except . and ..      -a  all files including .hidden
-b  escape nongraphic chars        -d  directory, not contents
-F  append /dir *exe @sym |FIFO    -f  files (no sort/filter/format)
-H  follow command line symlinks   -i  inode number
-L  follow symlinks                -N  no escaping, even on tty
-p  put '/' after dir names        -q  unprintable chars as '?'
-R  recursively list in subdirs    -s  storage used (in --block-size)
-Z  security context

output formats:
-1  list one file per line         -C  columns (sorted vertically)
-g  like -l but no owner           -h  human readable sizes
-l  long (show full details)       -ll long with nanoseconds (--full-time)
-m  comma separated                -n  long with numeric uid/gid
-o  long without group column      -r  reverse order
-w  set column width               -x  columns (horizontal sort)
```

```
apt install coreutils grep tar xz cpio gawk nc sed findutils patch
                                                                     ✔
2026-03-14  16:09.47  /home/mobaxterm  touch student.txt
                                                                     ✔
2026-03-14  17:41.10  /home/mobaxterm  mkdir data
                                                                     ✔
2026-03-14  17:41.19  /home/mobaxterm  cp student.txt data/
                                                                     ✔
2026-03-14  17:41.31  /home/mobaxterm  cd data
                                                                     ✔
2026-03-14  17:41.47  /home/mobaxterm/data  ls
student.txt
                                                                     ✔
2026-03-14  17:41.48  /home/mobaxterm/data  mv student.txt info.txt
                                                                     ✔
2026-03-14  17:41.56  /home/mobaxterm/data  ls
info.txt
                                                                     ✔
2026-03-14  17:41.57  /home/mobaxterm/data  rm info.txt
                                                                     ✔
2026-03-14  17:42.06  /home/mobaxterm/data  ls
                                                                     ✔
2026-03-14  17:42.07  /home/mobaxterm/data  rmdir data
rmdir: data: No such file or directory
                                                                     ✘
2026-03-14  17:42.16  /home/mobaxterm/data  cd..
cd..: command not found
                                                                     ✘
2026-03-14  17:42.25  /home/mobaxterm/data   cd ..
                                                                     ✔
2026-03-14  17:42.29  /home/mobaxterm  rmdir data
                                                                     ✔
2026-03-14  17:42.31  /home/mobaxterm  █
```

Assignment 02: File Editing & Text Processing

1. Create a file called bio.txt
Command :- touch bio.txt
I used the `touch` command to create a new empty file named **bio.txt**.

2. Add details using vi
Command: - vi bio.txt
I used the `vi` command to open the **bio.txt** file in the vi text editor and added the details name , course name and city
Type text in vi, I pressed **i** to enter insert mode and then typed the details.

3. Save the file properly
Command: - :wq
I used the `:wq` command in vi editor to save the file and exit the editor.

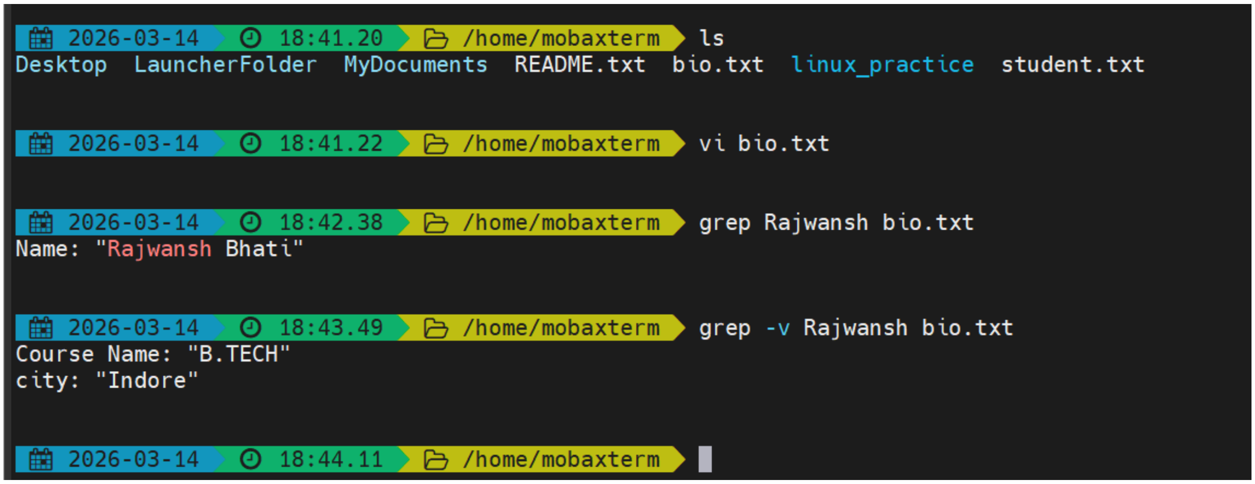
4. Search for your name inside the file
Command: - /Rajwansh
I used the `/` search command in vi editor to search for my name inside the **bio.txt** file.

5. Delete all content from the file using shortcut commands
Command: - gg and dG
I used the `gg` command to move the cursor to the beginning of the file and `dG` to delete all content from the file.

Using grep Command

6. Search your name in bio.txt
Command :- grep Rajwansh bio.txt
I used the `grep` command to search for my name in the **bio.txt** file.

7. Show lines that do NOT contain your name
Command: - grep -v Rajwansh bio.txt
I used the `grep -v` command to display the lines that do not contain my name in the bio.txt file.



```
2026-03-14  18:41.20  /home/mobaxterm  ls
Desktop  LauncherFolder  MyDocuments  README.txt  bio.txt  linux_practice  student.txt

2026-03-14  18:41.22  /home/mobaxterm  vi bio.txt

2026-03-14  18:42.38  /home/mobaxterm  grep Rajwansh bio.txt
Name: "Rajwansh Bhati"

2026-03-14  18:43.49  /home/mobaxterm  grep -v Rajwansh bio.txt
Course Name: "B.TECH"
city: "Indore"

2026-03-14  18:44.11  /home/mobaxterm
```

Assignment 03: Linux Permission

1. Create a file script.sh
Command : - touch script.sh
I used the `touch` command to create a new file named **script.sh**.

2. Check its permission using ls –l
Command : - ls -l script.sh
I used the `ls -l` command to check the file permissions of **script.sh**. This command shows the permission, owner, group, and file details.

Output:
-rw-rw-r-- 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh

3. Change permission to 755
Command : - chmod 755 script.sh
I used the chmod command to change the file permission to **755**.

Output:-
-rwxr-xr-x 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh

4. Change permission to 764
Command: - chmod 764 script.sh
I used the chmod command to set the permission of the file to **764**.

Output: -rwxrwxr-x 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh

5. Explain what 764 means.

Meaning of Permission 764:

- **Owner (7)** → The owner of the file has read, write, and execute permissions means all permission
- **Group (6)** → The group members have read and write permissions, but cannot execute the file.
- **Others (4)** → All other users have read-only permission. They cannot modify or execute the file.
  The file owner has full access, the group can read and modify the file, and everyone else can only view the file.

6. Give execute permission using symbolic method
Command:- chmod +x script.sh
I used the symbolic method chmod +x to add execute permission to the file **script.sh** so that it can be executed as a script.

7. What is the meaning of r=4, w=2, x=1?
Linux uses the numbers **4, 2, and 1** to represent read, write, and execute permissions, and they are combined to form permission codes like 755, 764, 777

8. What does permission 777 mean? Is it safe? Why?
Permission **777** means:

- Owner → read, write, execute
- Group → read, write, execute
- Others → read, write, execute

This means everyone can read, modify, and execute the file.

Safety:
Permission **777 is** generally not safe because any user can change or delete the file. It can create security risks, especially on shared systems

```
vboxuser@rajwansh-vm:~$ touch script.sh
vboxuser@rajwansh-vm:~$ ls
 Ajay              Hello.txt                        Rishu
 Ashish            IdeaProjects                     Rishu.txt
 aws               images                           Script
 awscliv2.zip      Lol                              scriptname.sh
 Backup            minikube-latest.aarch64.rpm      script.sh
 backup.log        minikube_latest_amd64.deb        session.txt
 Backup_record     minikube_latest_arm64.deb        shiv
 Demo.txt          Music                            Shiva
 Desktop           Neha                             snap
 Devops            nohup.out                        Sr.sh
 Documents         Pictures                         s.sh
 Downloads         Postman                          Swarnim
 emartapp          Practice                         Templates
 file1.txt         Print.txt                        timestamps.txt
 file2.txt         Public                           timestamp.txt
 file.txt          Python                           tree_2.0.2-1_arm64.deb
 Harsh             Raj                              typescript
 Harsh.txt        'Raj'$'\n'                        Videos
 health.sh         Raj.sh                           view
 hello.txt         Rajvir.txt
vboxuser@rajwansh-vm:~$ ls -l script.sh
-rw-rw-r-- 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh
vboxuser@rajwansh-vm:~$ chmod 755 script.sh
vboxuser@rajwansh-vm:~$ ls -l script.sh
-rwxr-xr-x 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh
vboxuser@rajwansh-vm:~$ chmod 764 script.sh
vboxuser@rajwansh-vm:~$ ls -l script.sh
-rwxrw-r-- 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh
vboxuser@rajwansh-vm:~$ chmod +x script.sh
vboxuser@rajwansh-vm:~$ ls -l script.sh
-rwxrwxr-x 1 vboxuser vboxuser 0 Mar 14 13:46 script.sh
```

Assignment 04: Linux Networking

Task 01 : Check Network Configuration

1.Check Your System IP Address
Command: - ifconfig
I used the ifconfig command to check the IP address of my system. This command displays network information such as the IP address, network interfaces, and their status. From the output, I was able to see the IP address assigned to my system.

Output:
inet addr:192.168.1.37 - This indicates that **192.168.1.37** is the IP address assigned to my system

2 . IP Address
An IP Address is a unique number assigned to a device connected to a network. It helps identify

and communicate with devices on a network or the internet. Every device such as a computer, phone, or server needs an IP address to send and receive data.

Task 02 : Test Network Reachability

1. Ping google.com
I used the ping command to check the connectivity between my system and google.com.

Output: -
Pinging google.com [142.250.206.110] with 32 bytes of data:
Reply from 142.250.206.110: bytes=32 time=8ms TTL=119
Reply from 142.250.206.110: bytes=32 time=7ms TTL=119
Reply from 142.250.206.110: bytes=32 time=6ms TTL=119
Reply from 142.250.206.110: bytes=32 time=8ms TTL=119
Ping statistics for 142.250.206.110:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 8ms, Average = 7ms

2. Observation
When I executed the ping google.com command, the system resolved the domain name google.com to the IP address 142.250.206.110
This means the DNS server successfully converted the domain name into its corresponding IP address.

Packet Loss % -: The ping showed Packets Sent: 4, Packets Received: 4, Packets Lost: 0, Packet Loss: 0%
This indicates that the network connection between my system and the server is working properly without any packet loss.

Task 03 : Check Port Connectivity

1. Check if port 80 is open on google.com using telnet
Command : - telnet google.com 80
I used the telnet command to check if port 80 is open on google.com. If the connection is successful, it means that Google's web server is running and accepting connections on that port, so users can access the website through a web browser.

2. What does it mean if Connection successful?

If the connection is successful, it means port 80 is open and the server is accepting HTTP requests. This indicates that the web service is running and accessible.

3. What does it mean if Connection refused?

If the connection is refused, it means the server is reachable but port 80 is closed or the service is not running on that port.

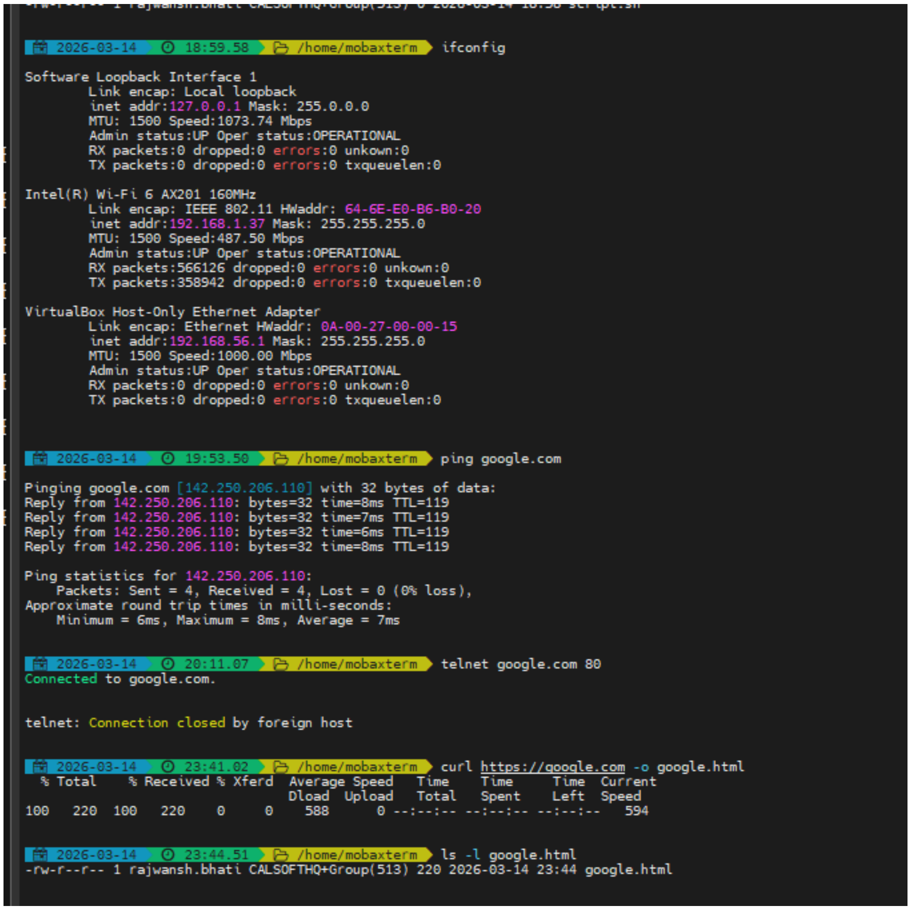
4. What does it mean if Connection timed out?

If the connection timed out, it means the system could not reach the server or the port is blocked by a firewall or network restriction.

Task 04: - Download data using Curl

1. Save it into a file.
Download google.com homepage
Command: - curl https://google.com -o google.html
I used the `curl` command to download the homepage of google.com and save it into a file named **google.html**.

2. Verify the file is created.
Command: - ls -l google.html
I used the `ls` command to verify that the file **google.html** was successfully created in the current directory.



Assignment 5: Process & System Monitoring

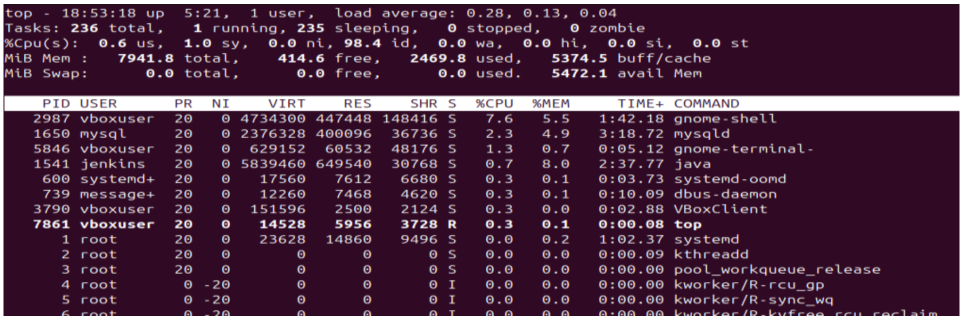
1. List all running processes.

Command: - ps –e

I used the `ps -e` command to list all the running processes in the system. This command displays the process ID (PID), terminal, time, and command for each running process.

2. Open top command and monitor CPU usage.

I used the `top` command to monitor the system processes in real time. It displays information such as CPU usage, memory usage, running processes, and system performance.



3. Check disk usage

Command: - df –h

I used the `df -h` command to check the disk usage of the system. It shows the available and used disk space of all mounted file systems.

4. Check memory usage

Command: - free –h

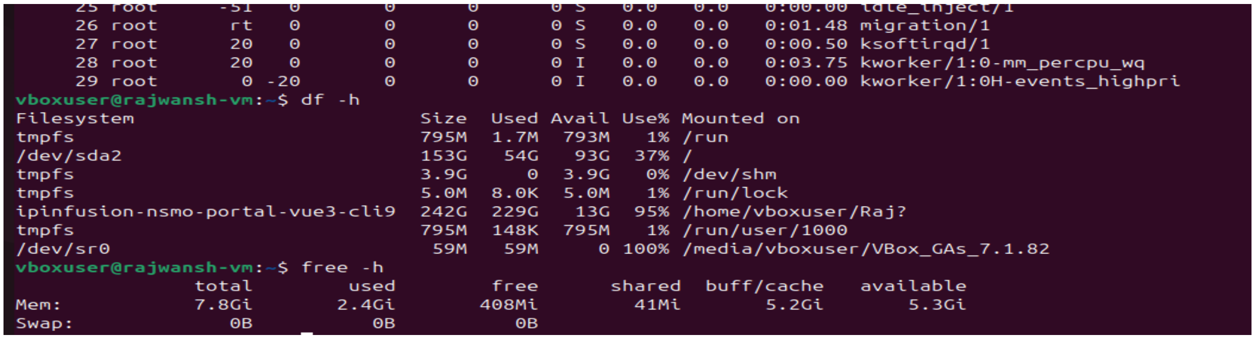
I used the `free -h` command to check the memory usage of the system. It displays total, used, and free memory in a human-readable format.

5. Difference between ps and top ?

PS	TOP
Shows a snapshot of running processes	Shows processes in real-time
Output is static (does not update automatically)	Output continuously updates
Used for quick process listing	Used for monitoring CPU and memory usage
Shows limited process information	Shows detailed system performance information

6. What does df -h show?

The `df -h` command displays the disk space usage of file systems in a human-readable format. It shows information such as total disk space, used space, available space, and the mount point of each file system.



```
     25 root      -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     26 root       rt   0       0      0      0 S   0.0   0.0   0:01.48 migration/1
     27 root       20   0       0      0      0 S   0.0   0.0   0:00.50 ksoftirqd/1
     28 root       20   0       0      0      0 I   0.0   0.0   0:03.75 kworker/1:0-mm_percpu_wq
     29 root        0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
vboxuser@rajwansh-vm:~$ df -h
Filesystem                        Size  Used Avail Use% Mounted on
tmpfs                             795M  1.7M  793M   1% /run
/dev/sda2                         153G   54G   93G  37% /
tmpfs                             3.9G     0  3.9G   0% /dev/shm
tmpfs                             5.0M  8.0K  5.0M   1% /run/lock
ipinfusion-nsmo-portal-vue3-cli9  242G  229G   13G  95% /home/vboxuser/Raj?
tmpfs                             795M  148K  795M   1% /run/user/1000
/dev/sr0                           59M   59M     0 100% /media/vboxuser/VBox_GAs_7.1.82
vboxuser@rajwansh-vm:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.8Gi       2.4Gi       408Mi        41Mi       5.2Gi       5.3Gi
Swap:             0B          0B          0B
```

Assignment 6: Shell Scripting

Task 1: Simple Script

Create Script File
Command: touch script1.sh

Script Code

#!/bin/bash

echo "Welcome to Linux Training"
date
whoami

I used the `echo` command to print the message **"Welcome to Linux Training"** on the screen.
I used the `date` command to display the **current date and time** of the system.
I used the `whoami` command to display the **current logged-in user name**.

Task 2: User Input Script

Create Script File
Command: - touch script2.sh

Script Code
#!/bin/bash
echo "Enter your name:"
read name
echo "Hello $name, welcome to Linux"

I used the `read` command to take the **user name as input** from the terminal.
The value entered by the user is stored in the variable **name**.
I used the `echo` command to display the message **"Hello <name>, welcome to Linux"**.

Conditional & Loop Script

Task 3: Even or Odd Program

Create Script File
Command: - touch script3.sh

Script Code

```
#!/bin/bash
echo "Enter a number:"
read num
if [ $num -lt 10 ]
then
echo "The number is less than 10"
else
echo "The number is greater than or equal to 10"
fi
```

I used the read command to take a number as input from the user.
The if condition checks whether the number is less than 10 using -lt (less than operator).
If the condition is true, it prints "The number is less than 10", otherwise it prints the other message.

Task 4: Loop Program

Create Script File
Command: - touch script4.sh

Script Code:

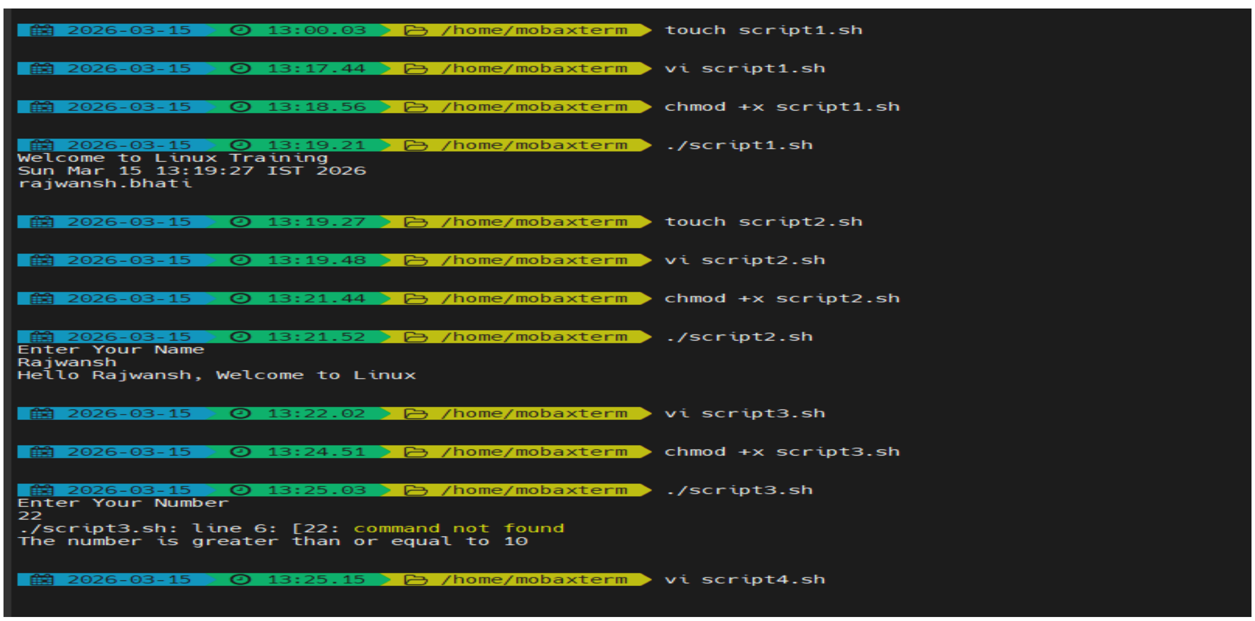
```
#!/bin/bash
for i in {1..10}
do
echo $i
done
```

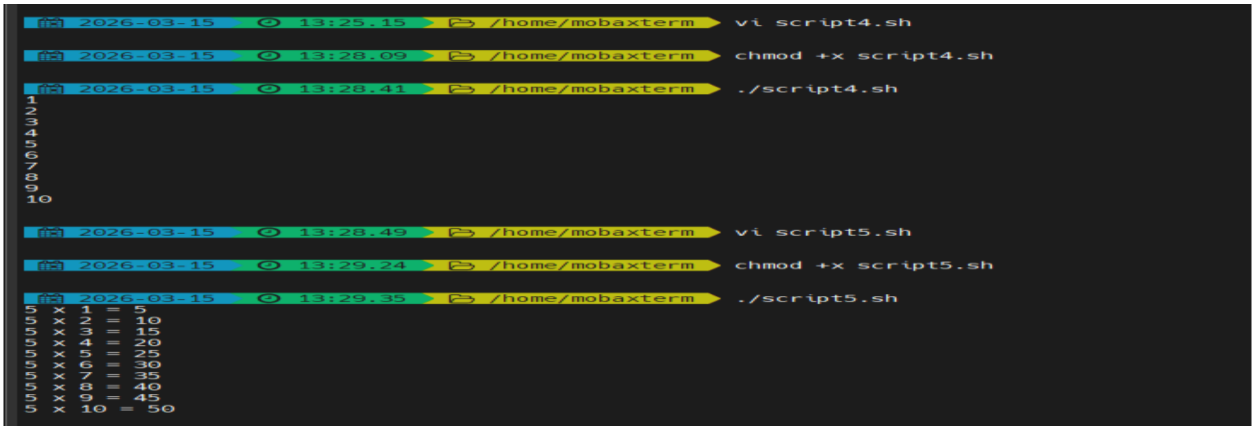
I used a for loop to print numbers from 1 to 10.
The loop runs from 1 to 10 and prints each number using the echo command.

Script Code of Multiplication table

```
#!/bin/bash
for i in {1..10}
do
echo "5 x $i = $((5*i))"
done
```

I used a for loop to generate the multiplication table of 5.
The loop runs from 1 to 10. $((5*i)) is used to perform multiplication and display the result.





Assignment 7 : Mini Project

Task: Backup Script

Command : - touch backup.sh
First Create a Backup.sh file because in Backup.sh I Write a Script code

Script Code

#!/bin/bash
if [-d backup]
then
echo "Directory already exists"
else
mkdir backup
fi
echo "Current date and time:"
date
cp *.txt backup/
echo "Backup Completed"

I used an if condition to check whether the backup directory already exists. If the directory exists, the script prints "Directory already exists". If the directory does not exist, it creates the backup directory using the mkdir command. I used the date command to display the current date and time. The cp command copies all .txt files into the backup folder. Finally, the script prints "Backup Completed".

```
 2026-03-15  13:47.50  /home/mobaxterm  touch backup.sh

 2026-03-15  13:47.59  /home/mobaxterm  vi backup.sh

 2026-03-15  13:48.54  /home/mobaxterm  chmod +x backup.sh

 2026-03-15  13:49.03  /home/mobaxterm  ./backup.sh
Current date and time:
Sun Mar 15 13:49:07 IST 2026
Backup Completed

 2026-03-15  13:49.07  /home/mobaxterm  ls
Desktop         MyDocuments  backup     bio.txt      hello.txt       script.sh    script2.sh  script4.sh  student.txt
LauncherFolder  README.txt   backup.sh  google.html  linux_practice  script1.sh   script3.sh  script5.sh

 2026-03-15  13:49.26  /home/mobaxterm  cd backup

 2026-03-15  13:49.32  /home/mobaxterm/backup  ls
README.txt  bio.txt  hello.txt  student.txt

 2026-03-15  13:49.37  /home/mobaxterm/backup  ./backup.sh
bash: ./backup.sh: No such file or directory

 2026-03-15  13:49.41  /home/mobaxterm/backup  cd ..

 2026-03-15  13:49.46  /home/mobaxterm  ./backup.sh
Directory already exists
Current date and time:
Sun Mar 15 13:49:49 IST 2026
Backup Completed
```